

PharmSignals

Drug Safety Signal Detection &
Regulatory Submission Readiness

TEAM:	Demon Slayer
PROBLEM:	P2
LICENSE:	Apache 2.0
TARGET:	Local Runtime [npm run dev]

The Human Cost of Data Overload

20,000,000+

reports in the FDA FAERS database.

Delayed disproportionality detection led to an estimated 27,000+ excess cardiovascular events before the Vioxx withdrawal.

The Financial Cost of Manual Audits

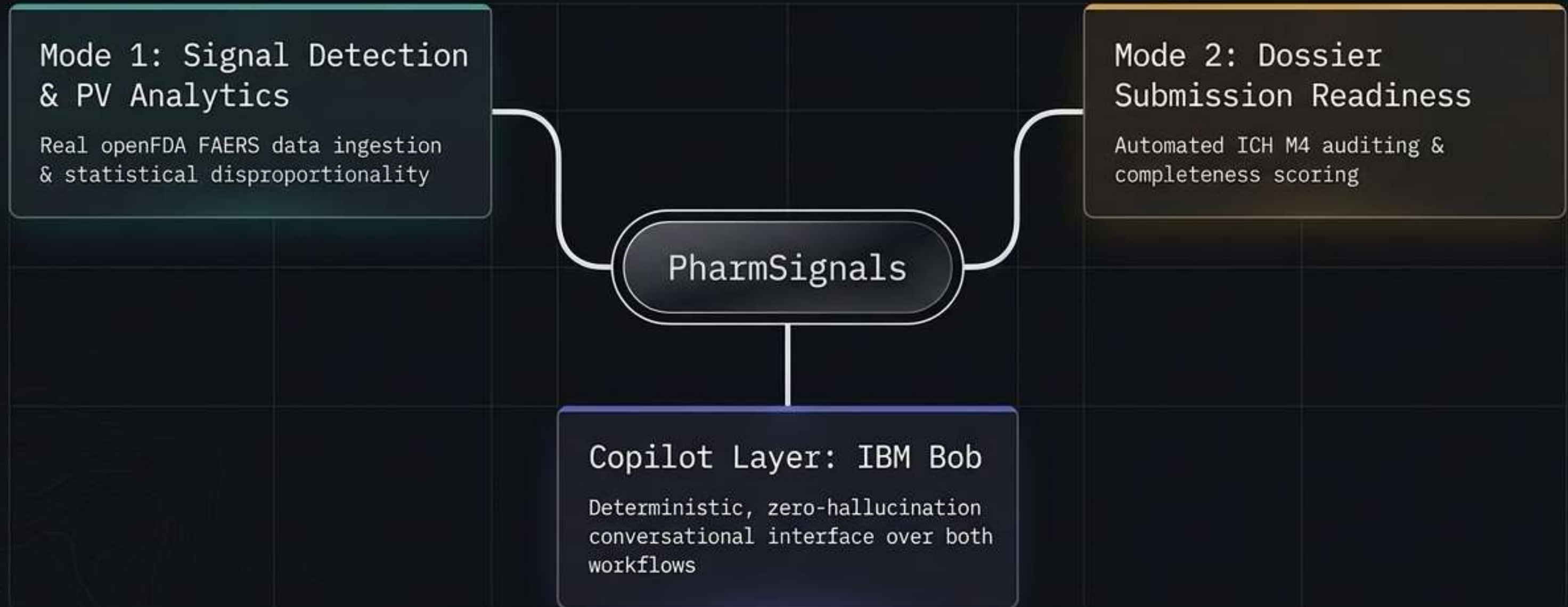
100,000+

page CTD dossiers spanning 5 ICH modules.

A single missing mandatory section causes a Refusal-to-File, delays delaying market entry by 6-12 months at a cost of \$50-100M.

Both domains fail because manual review cannot scale against complex, unstructured data volumes.

The Unified Workspace



A unified workspace replacing disconnected manual checklists with deterministic, statistical certainty.

System Architecture (Offline-First & Deterministic)



Frontend: Next.js 14, React 18, TypeScript, Tailwind.
(Single-page application with client-side tab switching).



API Gateway: FastAPI + Uvicorn.
(REST endpoint bridging the UI and the Python data science stack).



Processing Engines: Pandas, NumPy, SciPy, scikit-learn.
(In-memory execution for ephemeral, secure processing).



Data / File I/O: Local CSV/JSON/PDF ingest.

Core Engineering Philosophy

Deterministic Logic First.
Statistics and completeness scores are hardcoded mathematics.
AI is strictly confined to narration.

Mode 1 – Pharmacovigilance Statistical Engine

a	b
c	d



$$\text{Evans PRR} = \frac{[a/(a+b)]}{[c/(c+d)]}$$

+ Uncorrected Pearson Chi-Square
(`scipy.stats.chi2_contingency`)

Hardcoded Regulatory Thresholds

```
PRR >= 2.0  
Chi^2 >= 4.0  
min_cases (a) >= 3
```

Output: **SIGNAL** | **WEAK_SIGNAL** | **NOISE**

Ingest: Cleans and normalises real openFDA data (48,311 row benchmark subset).

Digital Twin Backtesting (The Vioxx Proof)



**+242 Days
Lead Time**

PharmSignals detected the signal approximately 8 months before the official 30 September 2004 market withdrawal, using a strict walk-forward monthly recomputation methodology.

Backtest Synthesis – Radical Transparency

Avandia

**+1,205 days
early detection**



Detected over 3 years before
the 2007 FDA boxed warning.

Vioxx

**+242 days
early detection**



Detected ~8 months before the
2004 market withdrawal.

Baycol (The Limitation)

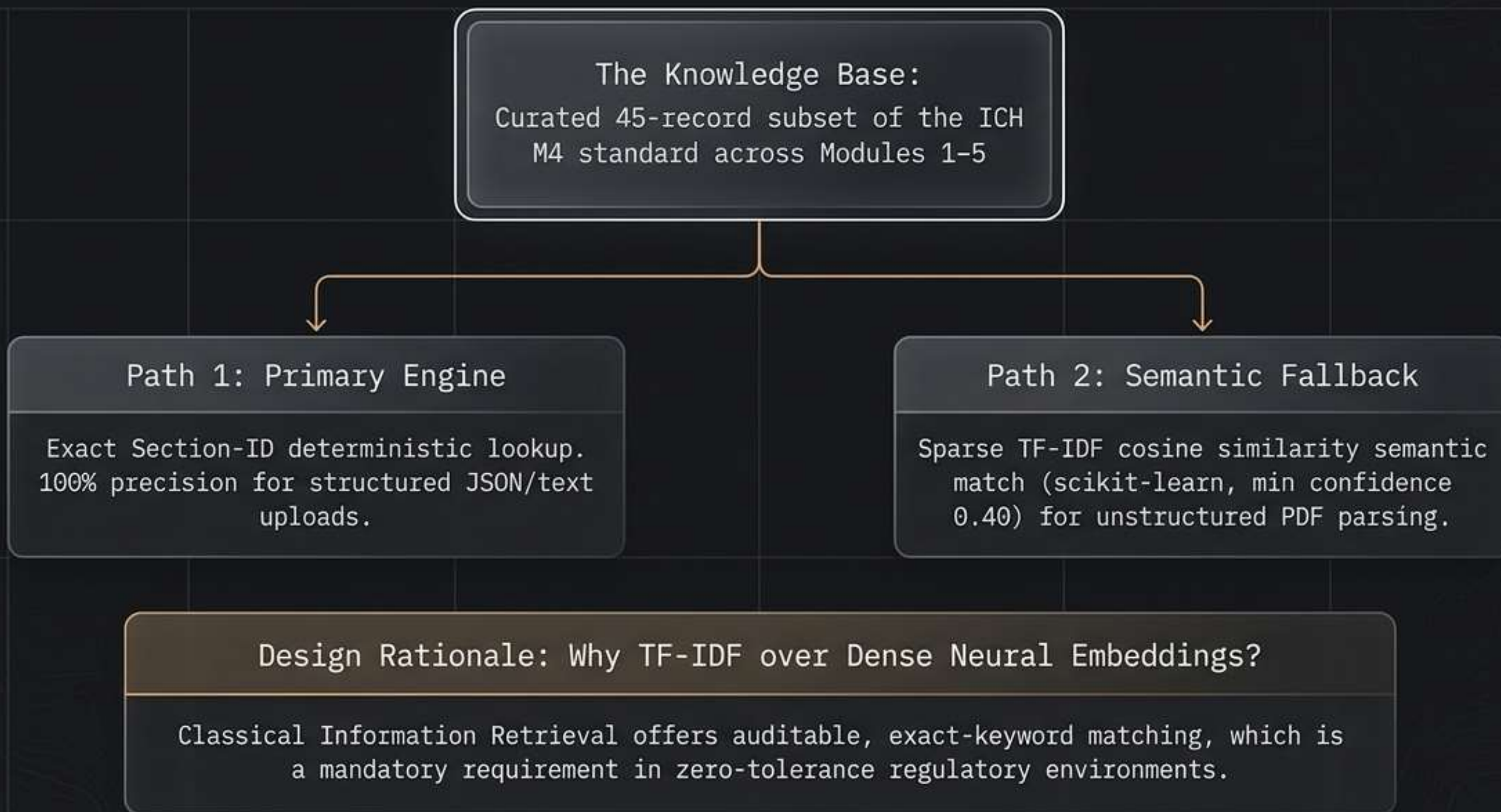
DATA_UNAVAILABLE_PRE_WITHDRAWAL



Withdrawn in August 2001. The
openFDA electronic records
begin circa 2004.

Radical Transparency in Healthcare Analytics. We explicitly flag historical data boundaries rather than fabricating backward projections. Limitations (like static background approximation) are proudly disclosed in the codebase.

Mode 2 – CTD Submission Readiness Audit



Deterministic Completeness Scoring

$$\text{Score} = \frac{(\text{Present} \times 1.0 + \text{Partial} \times 0.5 + \text{Missing} \times 0.0)}{\text{Total_Required} \times 100}$$

Detection Logic: Identifies 'Partial' gaps deterministically via regex searching for draft language (e.g., 'TBD', 'placeholder', 'pending').

GapReportOutput

Priority Gap Matrix

CRITICAL

- Immediate Remediation Required: Missing Section 3.2

MAJOR

- Address Placeholder: 'TBD' in Module 2.4

STANDARD

- Complete Pending Items: Draft Language Detected in 5.1

OPTIONAL

- Review Optional Elements: Consider Supplementing 1.6

AI Copilot – The ‘Zero Hallucination’ Layer

What are the latest signals for Avandia?

Step 1: The Strict Rule Engine (Primary)

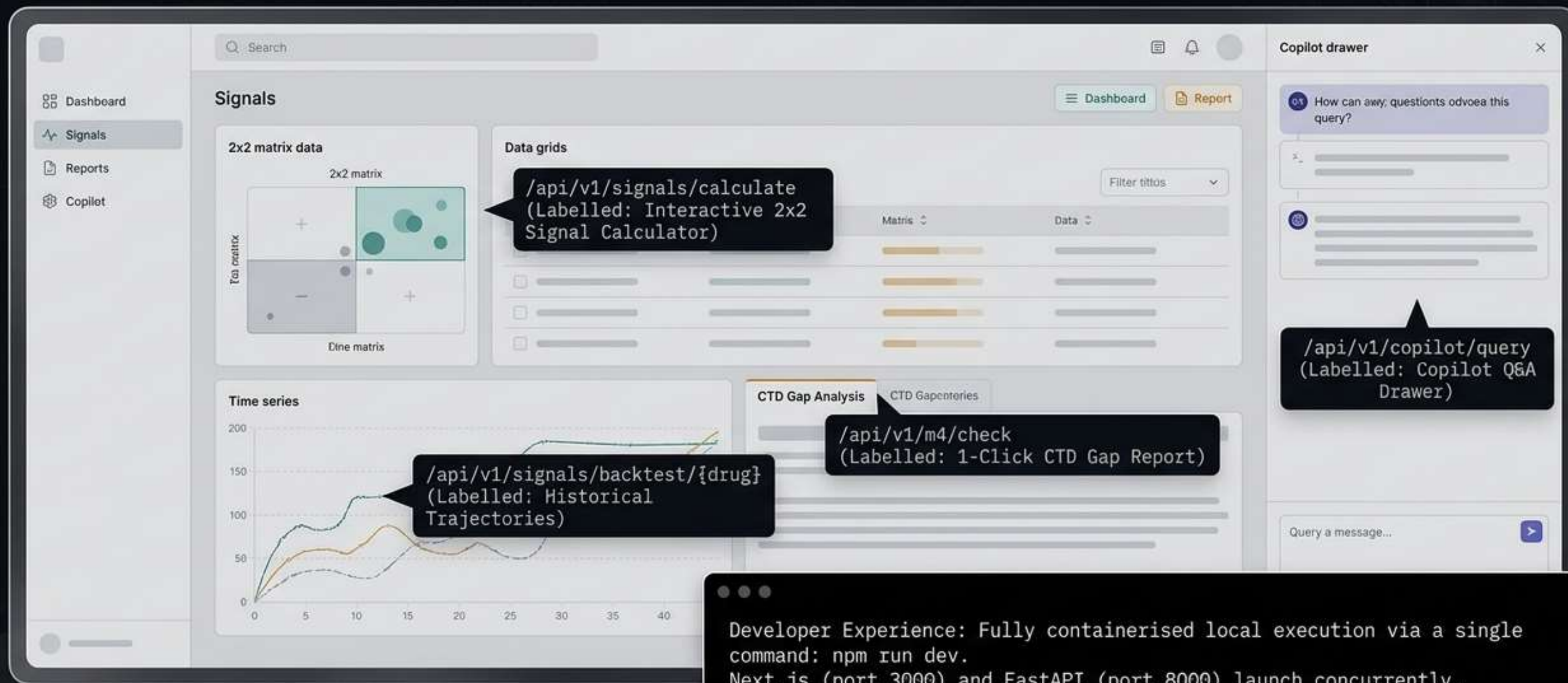
- Hand-written Python templates intercept the query.
- Pulls live, pre-computed PRR numbers and backtest data directly from the Pandas/SciPy pipeline.
- Result: Deterministic, zero-hallucination narrative generation. Fully offline.

Step 2: Gemini 2.5 Flash (Fallback)

Optional external REST API call for broader narrative reasoning, strictly grounded in the extracted statistical context.

Architectural Stance: AI is used strictly for conversational UX and routing, never for generating numerical safety data or completeness scores.

The Unified Workflow (Putting it together)



Engineering Integrity & Limitations

What We Didn't Do

No Black-Box LLM Scoring

No Fabricated Scale

Disabled AE Clustering

Offline-First Copilot

Why We Didn't Do It

Used verifiable Evans PRR and explicit ICH M4 completeness mathematics.

Limited to 3 benchmark drugs (48k rows) and 45 ICH M4 sections to ensure absolute accuracy in our testing environment.

Explicitly omitted this P2 requirement to guarantee zero compromises on the core PRR statistical pipeline.

Built deterministic rules over mocking watsonx APIs. The architecture is enterprise-ready for future IBM model injection.

Roadmap & Future Expansion

V1 (Current State)

- Disproportionality Signal Detection (PRR/Chi²).
- Core ICH M4 Auditing (TF-IDF hybrid RAG).
- ~100 automated test suite coverage (pytest).

V2 (Near Term)

- Full FAERS database scale-up (20M+ rows) unlocking the declared PostgreSQL architecture.
- Expansion to regional CTD variants (e.g., EMA standards).

V3 (Enterprise Future)

- Continuous, real-time PV data stream surveillance.
- Native IBM watsonx.ai integration for automated drafting of Risk Evaluation and Mitigation Strategies (REMS).

PharmSignals replaces manual checklist fatigue with statistical certainty.

Identifies known high-risk signal patterns months in advance.

Delivers structured, deterministic, and auditable CTD gap reporting.

Transforms pharmacovigilance and regulatory compliance into a unified, proactive science.